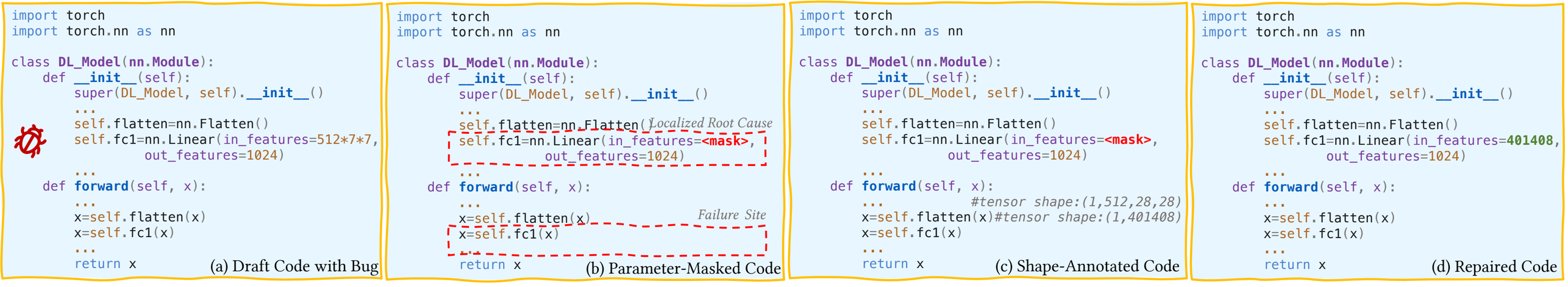
import torch
import torch.nn as nn
class DL_Model(nn.Module):
def __init__(self):
super(DL_Model, self).__init__()
...
self.flatten=nn.Flatten()
self.fc1=nn.Linear(in_features=512*7*7,
out_features=1024)
...
def forward(self, x):
...
x=self.flatten(x)
x=self.fc1(x)
...
return x
(a) Draft Code with Bug
import torch
import torch.nn as nn
class DL_Model(nn.Module):
def __init__(self):
super(DL_Model, self).__init__()
...
self.flatten=nn.Flatten()
Localized Root Cause
self.fc1=nn.Linear(in_features=<mask>,
out_features=1024)
...
def forward(self, x):
...
x=self.flatten(x)
Failure Site
x=self.fc1(x)
...
return x
(b) Parameter-Masked Code
import torch
import torch.nn as nn
class DL_Model(nn.Module):
def __init__(self):
super(DL_Model, self).__init__()
...
self.flatten=nn.Flatten()
self.fc1=nn.Linear(in_features=<mask>,
out_features=1024)
...
def forward(self, x):
...
#tensor shape:(1,512,28,28)
x=self.flatten(x)#tensor shape:(1,401408)
x=self.fc1(x)
...
return x
(c) Shape-Annotated Code
import torch
import torch.nn as nn
class DL_Model(nn.Module):
def __init__(self):
super(DL_Model, self).__init__()
...
self.flatten=nn.Flatten()
self.fc1=nn.Linear(in_features=401408,
out_features=1024)
...
def forward(self, x):
...
x=self.flatten(x)
x=self.fc1(x)
...
return x
(d) Repaired Code